

Statistics

0 Revision on IA Probability

Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is the sample space, $\mathcal{F}$ is the set of events, and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is the probability measure.

Definition 0.1 – Cumulative distribution function

$$F_X(x) = \mathbb{P}(X \leq x)$$

Definition 0.2 – Probability mass function

A discrete random variable takes values in a countable set $\mathcal{X}$ and has probability mass function (pmf)

$$p_X(x) = \mathbb{P}(X = x).$$

Definition 0.3 – Probability density function

A continuous random variable has probability density function (pdf) f_X satisfying

$$\mathbb{P}(X \in A) = \int_A f_X(x) dx$$

for nice sets A .

Definition 0.4 – Independence

We say that $X_1, \dots, X_n$ are independent if

$$\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) = \mathbb{P}(X_1 \leq x_1) \dots \mathbb{P}(X_n \leq x_n)$$

for all $(x_1, \dots, x_n)$. If $X_1, \dots, X_n$ have pdfs or pmfs $f_{X_1}, \dots, f_{X_n}$, this is equivalent to

$$f_X(x) = \prod_{i=1}^n f_{X_i}(x_i).$$

Definition 0.5 – Expectation

$$\mathbb{E}(X) = \begin{cases} \sum_{x \in \mathcal{X}} x p_X(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x f_X(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Definition 0.6 – Variance

$$\text{var}(X) = \mathbb{E}[(X - \mathbb{E}(X))^2]$$

Definition 0.7 – Moment generating function

$$M(t) = \mathbb{E}(e^{tX})$$

Note that

$$\mathbb{E}(X^n) = \left[\frac{d^n}{dt^n} M(t) \right]_{t=0}.$$

Under mild conditions, $M_X = M_Y$ implies $F_X = F_Y$.

If $a_1, \dots, a_n \in \mathbb{R}$, then

$$\begin{aligned} \mathbb{E}(a_1 X_1 + \dots + a_n X_n) &= a_1 \mathbb{E}(X_1) + \dots + a_n \mathbb{E}(X_n) \\ \text{var}(a_1 X_1 + \dots + a_n X_n) &= \sum_{i,j=1}^n a_i a_j \text{cov}(X_i, X_j) \end{aligned}$$

where $\text{cov}(X_i, X_j) = \mathbb{E}[(X_i - \mathbb{E}(X_i))(X_j - \mathbb{E}(X_j))]$. In vector notation,

$$\begin{aligned} \mathbb{E}(a^\top X) &= a^\top \mathbb{E}(X) \\ \text{var}(a^\top X) &= a^\top \text{var}(X) a \end{aligned}$$

where $\text{var}(X)$ is the covariance matrix of X and has entries $\text{cov}(X_i, X_j)$.

If X and Y are discrete random variables with joint pmf $p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$ and marginal pmf $p_Y(y) = \mathbb{P}(Y = y) = \sum_{x \in \mathcal{X}} p_{X,Y}(x, y)$, then the conditional pmf is

$$p_{X|Y}(x | y) = \mathbb{P}(X = x | Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

If X and Y are continuous random variables, the joint pdf $f_{X,Y}$ satisfies $\mathbb{P}(X = x, Y = y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(x, y) dx dy$ and the marginal pdf of Y is $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$. The conditional pdf of X given Y is

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Definition 0.8 – Conditional expectation

$$\mathbb{E}(X | Y) = \begin{cases} \sum_{x \in \mathcal{X}} x p_{X|Y}(x | Y) & \text{for discrete} \\ \int_{-\infty}^{\infty} x f_{X|Y}(x | Y) dx & \text{for continuous} \end{cases}$$

Note that $\mathbb{E}(X | Y)$ is a function of Y so it is a random variable. The tower property of expectation is

$$\mathbb{E}(X) = \mathbb{E}(\mathbb{E}(X | Y)).$$

This is a consequence of the law of total probability

$$p_X(x) = \sum_{y \in \mathcal{Y}} p_{X|Y}(x | y) p_Y(y).$$

Proposition 0.1 – Law of total variance

$$\text{var}(X) = \mathbb{E}[\text{var}(X | Y)] + \text{var}(\mathbb{E}(X | Y))$$

Proof.

$$\begin{aligned}
\text{var}(X) &= \mathbb{E}(X^2) - \mathbb{E}(X)^2 \\
&= \mathbb{E}(\mathbb{E}(X^2 | Y)) - \mathbb{E}(\mathbb{E}(X | Y))^2 \\
&= \mathbb{E}[\underbrace{\mathbb{E}(X^2 | Y) - \mathbb{E}(X | Y)^2}_{\text{var}(X | Y)}] + \mathbb{E}(\mathbb{E}(X | Y)^2) - \mathbb{E}(\mathbb{E}(X | Y))^2 \\
&= \mathbb{E}[\text{var}(X | Y)] + \text{var}(\mathbb{E}(X | Y))
\end{aligned}$$

□

Let $(x, y) \mapsto (u, v)$ be a differentiable bijection $\mathbb{R}^2 \rightarrow \mathbb{R}^2$. Then,

$$f_{U,V}(u, v) = f_{X,Y}(x(u, v), y(u, v)) |\det(J)|$$

where $J = \partial(x, y) / \partial(u, v)$ is the Jacobian.

Suppose $X_1, \dots, X_n$ are independent and identically distributed random variables with $\mathbb{E}(X_1) = \mu$ and $\text{var}(X_1) = \sigma^2$. Define the sum $S_n = \sum_{i=1}^n X_i$ and the sample mean $\bar{X}_n = S_n/n$. The weak law of large numbers states $\bar{X}_n \rightarrow \mu$ in probability, meaning that $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$ for all $\varepsilon > 0$. The strong law of large numbers states $\bar{X}_n \rightarrow \mu$ almost surely, meaning that $\mathbb{P}(\lim_{n \rightarrow \infty} \bar{X}_n = \mu) = 1$.

Theorem 0.1 – Central limit theorem

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} \approx \mathcal{N}(0, 1)$$

for large n . Colloquially, $S_n \approx \mathcal{N}(n\mu, n\sigma^2)$. Precisely, $\mathbb{P}(Z_n \leq z) \rightarrow \Phi(z)$ in distribution for all $z \in \mathbb{R}$, where Φ is the cdf of $\mathcal{N}(0, 1)$.

1 Parametric estimation

Suppose we observed data $X_1, \dots, X_n$ which are independent and identically distributed (iid) random variables taking values in $\mathcal{X}$. Let $X = (X_1, \dots, X_n)$. We assume that X_1 belongs to a statistic model $\{p(x, \theta) : \theta \in \Theta\}$ where θ is an unknown parameter. For example, we could have $X_1 \sim \text{Poisson}(\lambda)$ where $\theta = \lambda \in \Theta = (0, \infty)$. We might also have $X_1 \sim \mathcal{N}(\mu, \sigma^2)$ where $\theta = (\mu, \sigma^2) \in \Theta = \mathbb{R} \times (0, \infty)$. We would want to

- (i) Give an estimate $\hat{\theta} : \mathcal{X}^n \rightarrow \Theta$ of the true value of θ .
- (ii) Give an interval estimator $(\hat{\theta}_1(X), \hat{\theta}_2(X))$ of θ .
- (iii) Test some hypothesis about θ , for example, $H_0 : \theta = 0$.

Definition 1.1 – Estimator and sampling distribution

An estimator or statistic is a function of a data $T(X) = \hat{\theta}$ which we use to approximate the true parameter θ . The distribution of $T(X)$ is the sampling distribution.

Example 1.1

Suppose $X_1, \dots, X_n$ are iid with $X_1 \sim \mathcal{N}(\mu, 1)$. Let

$$\hat{\mu} = T(X) = \frac{1}{n} \sum_{i=1}^n X_i.$$

The sampling distribution of μ is $T(X) \sim \mathcal{N}(\mu, 1/n)$.

Definition 1.2 – Bias

The bias of a random variable $\hat{\theta} = T(X)$ is

$$\text{bias}(\hat{\theta}) = \mathbb{E}_{\theta}(\hat{\theta}) - \theta$$

where the expected value is taken over the model $X_i \sim f_X(\cdot | \theta)$.

In general, $\text{bias}(\hat{\theta})$ might be a function of θ , which is not explicit in notation.

Definition 1.3 – Unbiased estimator

We say that $\hat{\theta}$ is unbiased if $\text{bias}(\hat{\theta}) = 0$ for all $\theta \in \Theta$.

The estimator $\hat{\mu}$ from Example 1.1 is unbiased for μ since

$$\mathbb{E}_{\mu}(\hat{\mu}) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mu}(X_i) = \mu.$$

Definition 1.4 – Mean squared error

The mean squared error (mse) of an estimator $\hat{\theta}$ is

$$\text{mse}(\hat{\theta}) = \mathbb{E}_{\theta}[(\hat{\theta} - \theta)^2].$$

Again, $\text{mse}(\hat{\theta})$ is generally a function of θ .

Proposition 1.1 – Bias-variance decomposition

$$\text{mse}(\hat{\theta}) = \text{bias}(\hat{\theta})^2 + \text{var}_{\theta}(\hat{\theta})$$

Proof.

$$\begin{aligned} \text{mse}(\hat{\theta}) &= \mathbb{E}_{\theta}[(\hat{\theta} - \theta)^2] \\ &= \mathbb{E}_{\theta}[(\hat{\theta} - \mathbb{E}_{\theta}(\hat{\theta}) + \mathbb{E}_{\theta}(\hat{\theta}) - \theta)^2] \\ &= \underbrace{\mathbb{E}_{\theta}[(\hat{\theta} - \mathbb{E}_{\theta}(\hat{\theta}))^2]}_{=\text{var}_{\theta}(\hat{\theta})} + \underbrace{(\mathbb{E}_{\theta}(\hat{\theta}) - \theta)^2}_{=\text{bias}(\hat{\theta})^2} + 2(\mathbb{E}_{\theta}(\hat{\theta}) - \theta) \underbrace{\mathbb{E}_{\theta}[\hat{\theta} - \mathbb{E}_{\theta}(\hat{\theta})]}_{=0} \end{aligned}$$

□

Example 1.2

Suppose $X \sim \text{Binomial}(n, \theta)$ where n is known. We want to estimate $\theta \in [0, 1]$. The standard estimate is $T_u = X/n$. $\mathbb{E}_{\theta}(T_u) = \mathbb{E}(X)/n = (n\theta)/n = \theta$ so T_u is unbiased.

$$\text{mse}(T_u) = \text{var}_{\theta}(T_u) + \text{bias}(T_u)^2 = \frac{1}{n^2} \text{var}_{\theta}(X) = \frac{n\theta(1-\theta)}{n^2} = \frac{\theta(1-\theta)}{n}.$$

Now consider $T_b = (X + 1)/(n + 2)$. We can think of this as a weighted average $T_b = w(X/n) + (1 -$

$w)(1/2)$ where $w = n/(n+2)$.

$$\begin{aligned}\text{bias}(T_b) &= \mathbb{E}_\theta(T_b) - \theta = \frac{\mathbb{E}(X) + 1}{n+2} - \theta = (1-w) \left(\frac{1}{2} - \theta \right) \\ \text{var}(T_b) &= \frac{\text{var}_\theta(X)}{(n+2)^2} = \dots = w^2 \frac{\theta(1-\theta)}{n} \\ \text{mse}(T_b) &= w^2 \underbrace{\frac{\theta(1-\theta)}{n}}_{=\text{mse}(T_u)} + (1-w)^2 \left(\frac{1}{2} - \theta \right)^2\end{aligned}$$

We see that the biased estimator is not necessarily worse than the unbiased one in terms of mse.

Suppose $X_1, \dots, X_n$ are iid random variables taking values in $\mathcal{X}$, with pdf (or pmf) $f_X(\cdot | \theta)$. Consider θ as fixed and denote $X = (X_1, \dots, X_n)$.

Definition 1.5 – Sufficient statistic

A statistic T is sufficient for θ if the conditional distribution of X given $T(X)$ does not depend on θ .

Note that the parameter θ may be a vector, and $T(X)$ may also be a vector.

Example 1.3

Suppose $X_1, \dots, X_n \sim \text{Bernoulli}(\theta)$ for some $\theta \in [0, 1]$ are iid. Then,

$$f_X(x | \theta) = \prod_{i=1}^n \theta^{x_i} (1-\theta)^{1-x_i} = \theta^{\sum x_i} (1-\theta)^{n-\sum x_i}.$$

Define $T(X) = \sum_{i=1}^n x_i$. We can check that

$$f_{X|T=t}(x | T(x) = t) = \frac{\mathbb{P}_\theta(X = x, T(x) = t)}{\mathbb{P}_\theta(T(x) = t)} = \frac{\mathbb{P}_\theta(X = x)}{\mathbb{P}_\theta(T(x) = t)} = \frac{\theta^{\sum x_i} (1-\theta)^{n-\sum x_i}}{\binom{n}{t} \theta^t (1-\theta)^{n-t}} = \frac{1}{\binom{n}{t}}$$

which does not depend on θ . Hence, T is sufficient.

Theorem 1.1 – Factorisation criterion

The statistic T is sufficient for θ if and only if $f_X(x | \theta) = g(T(x), \theta)h(x)$ for some suitable functions g and h .

Proof. Suppose $f_{X|\theta}(x | \theta) = g(T(x), \theta)h(x)$. Compute

$$f_{X|T=t}(x | T = t) = \frac{\mathbb{P}_\theta(X = x, T(x) = t)}{\mathbb{P}_\theta(T(x) = t)} = \frac{g(T(x), \theta)h(x)}{\sum_{\{x' \in \mathcal{X} : T(x')=t\}} g(t, \theta)h(x')} = \frac{h(x)}{\sum_{\{x' \in \mathcal{X} : T(x')=t\}} h(x')}$$

which does not depend on θ so T is sufficient. Conversely, suppose T is sufficient. We can write

$$\mathbb{P}_\theta(X = x) = \mathbb{P}_\theta(X = x, T(X) = T(x)) = \underbrace{\mathbb{P}_\theta(X = x | T(X) = T(x))}_{h(x) \text{ since it does not depend on } \theta} \underbrace{\mathbb{P}_\theta(T(X) = T(x))}_{g(T(x), \theta)}.$$

□

In the case of Example 1.3, $g(t, \theta) = \theta^t (1-\theta)^{n-t}$ and $h(x) = 1$.

Example 1.4

Let $X_1, \dots, X_n \sim \text{Uniform}[0, \theta]$ for some $\theta \in (0, \infty)$ be iid.

$$f_X(x | \theta) = \prod_{i=1}^n \frac{1}{\theta} \mathbb{1}\{x_i \in [0, \theta]\} = \frac{1}{\theta^n} \underbrace{\mathbb{1}\{\max_i x_i \leq \theta\}}_{g(T(x), \theta)} \underbrace{\mathbb{1}\{\min_i x_i \geq 0\}}_{h(x)}$$

so $T(X) = \max_i X_i$ is a sufficient statistic.

Definition 1.6 – Minimal sufficient statistic

A sufficient statistic $T(X)$ is minimal sufficient if it is a function of every other sufficient statistic. That is, if $T'(X)$ is also sufficient, then $T'(x) = T'(y) \implies T(x) = T(y)$ for all $x, y \in \mathcal{X}$.

Minimal sufficient statistics are unique up to bijection. If T and T' are both minimal sufficient, then $T(x) = T(y) \iff T'(x) = T'(y)$.

Theorem 1.2

Suppose $T(X)$ is a statistic such that $f_X(x | \theta) / f_X(y | \theta)$ is constant as a function of θ if and only if $T(x) = T(y)$. Then, T is minimal sufficient.

Proof. We need to show that such a statistic is sufficient and minimal. Let z_t be a representative element of $\{x : T(x) = t\}$. We write

$$f_X(x | \theta) = \underbrace{f_X(z_{T(x)} | \theta)}_{g(T(x), \theta)} \underbrace{f_X(z_{T(x)} | \theta)}_{h(x)}$$

and use the factorisation theorem. Thus, T is sufficient. We now show that T is minimal as well. Let S be any other sufficient statistic. By the factorisation criterion, there exist g_S and h_S such that $f_X(x | \theta) = g_S(S(x), \theta) h_S(x)$. Suppose $S(x) = S(y)$, then

$$\frac{f_X(x | \theta)}{f_X(y | \theta)} = \frac{g_S(S(x), \theta) h_S(x)}{g_S(S(y), \theta) h_S(y)} = \frac{h_S(x)}{h_S(y)}$$

which does not depend on θ . By the hypothesis, $T(x) = T(y)$ so T is minimal sufficient. $\square$

Example 1.5

Suppose $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ are iid. Then,

$$\begin{aligned} \frac{f_X(x | \mu, \sigma^2)}{f_X(y | \mu, \sigma^2)} &= \frac{(2\pi\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2 \right]}{(2\pi\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} \sum_i (y_i - \mu)^2 \right]} \\ &= \exp \left[-\frac{1}{2\sigma^2} \left(\sum_i x_i^2 - \sum_i y_i^2 \right) + \frac{\mu}{\sigma^2} \left(\sum_i x_i - \sum_i y_i \right) \right]. \end{aligned}$$

This is constant in (μ, σ^2) if and only if $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$ and $\sum_{i=1}^n x_i^2 = \sum_{i=1}^n y_i^2$. Hence, $T(X) = (\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2)$ is minimal sufficient. Recall that bijections of minimal sufficient statistics are also minimal sufficient, so we can conclude that $S(X) = (\bar{X}, S_{XX})$ where $\bar{X} = (1/n) \sum_{i=1}^n X_i$ and $S_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2$ is also minimal sufficient.

Until now, we used $\mathbb{E}_\theta$ and $\mathbb{P}_\theta$ to denote the expectation and probability when $X_1, \dots, X_n$ are iid from a distribution with pdf (or pmf) $f_X(x | \theta)$. From now on, we drop the subscript θ to simplify notation.

Theorem 1.3 – Rao-Blackwell theorem

Let T be a sufficient statistic for θ and let $\tilde{\theta}$ be an estimator for θ with $\mathbb{E}(\tilde{\theta}^2) < \infty$ for all θ . Define a new estimator $\hat{\theta} = \mathbb{E}(\tilde{\theta} | T(X))$. Then for all θ ,

$$\mathbb{E}[(\hat{\theta} - \theta)^2] \leq \mathbb{E}[(\tilde{\theta} - \theta)^2].$$

The inequality is strict unless $\tilde{\theta}$ is a function of $T(X)$.

Proof. By the tower property of expectation, $\mathbb{E}(\hat{\theta}) = \mathbb{E}[\mathbb{E}(\tilde{\theta} | T)] = \mathbb{E}(\tilde{\theta})$ so $\text{bias}(\hat{\theta}) = \text{bias}(\tilde{\theta})$ for all θ . By the conditional variance formula,

$$\text{var}(\tilde{\theta}) = \mathbb{E}[\text{var}(\tilde{\theta} | T)] + \text{var}(\mathbb{E}(\tilde{\theta} | T)) = \mathbb{E}[\text{var}(\tilde{\theta} | T)] + \text{var}(\hat{\theta}) \geq \text{var}(\hat{\theta})$$

so $\text{mse}(\tilde{\theta}) \geq \text{mse}(\hat{\theta})$. Equality is achieved only when $\text{var}(\tilde{\theta} | T) = 0$ with probability one, which requires $\tilde{\theta}$ to be a function of T . $\square$

Note that $\hat{\theta}(T) = \int \tilde{\theta}(x) f_{X|T}(x | T) dx$. By the sufficiency of T , the conditional pdf does not depend on θ , so $\hat{\theta}$ does not depend on θ , and is a valid estimator.

Example 1.6

Suppose $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$ are iid. Let $\theta = \mathbb{P}(X_1 = 0) = e^{-\lambda}$.

$$f_X(x | \lambda) = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod x_i!} = \frac{\theta^n (-\log \theta)^{\sum x_i}}{\prod x_i!}$$

By the factorisation criterion, $T(X) = \sum_{i=1}^n x_i$ is sufficient. Recall that $\sum_{i=1}^n X_i \sim \text{Poisson}(n\lambda)$. Let $\tilde{\theta} = \mathbb{1}(X_1 = 0)$. Then,

$$\begin{aligned} \hat{\theta} &= \mathbb{E}(\tilde{\theta} | T = t) = \mathbb{P}(X_1 = 0 | \sum_{i=1}^n X_i = t) \\ &= \frac{\mathbb{P}(X_1 = 0, \sum_{i=2}^n X_i = t)}{\mathbb{P}(\sum_{i=1}^n X_i = t)} \\ &= \frac{\mathbb{P}(X_1 = 0) \mathbb{P}(\sum_{i=2}^n X_i = t)}{\mathbb{P}(\sum_{i=1}^n X_i = t)} \\ &= \frac{e^{-\lambda} e^{-(n-1)\lambda} ((n-1)\lambda)^t / t!}{e^{-n\lambda} (n\lambda)^t / t!} = \left(\frac{n-1}{n} \right)^t. \end{aligned}$$

Hence, $\hat{\theta} = (1 - 1/n)^{\sum x_i}$ has $\text{mse}(\hat{\theta}) < \text{mse}(\tilde{\theta})$ for all θ . As a sanity check, $\hat{\theta} = (1 - 1/n)^{n\bar{X}} \rightarrow e^{-\bar{X}} \rightarrow e^{-\lambda} = \theta$ as $n \rightarrow \infty$.

Example 1.7

Let $X_1, \dots, X_n \sim \text{Uniform}[0, \theta]$ are iid and suppose we want to estimate $\theta \geq 0$. In Example 1.4, we saw that $T = \max_i X_i$ is sufficient for θ . Let $\tilde{\theta} = 2X_1$ be an unbiased estimator.

$$\begin{aligned} \hat{\theta} &= \mathbb{E}(\tilde{\theta} | T = t) = 2\mathbb{E}(X_1 | \max X_i = t) \\ &= 2\mathbb{E}(X_1 | \max X_i = t, X_1 = \max X_i) \mathbb{P}(X_1 = \max X_i | \max X_i = t) \\ &\quad + 2\mathbb{E}(X_1 | \max X_i = t, X_1 \neq \max X_i) \mathbb{P}(X_1 \neq \max X_i | \max X_i = t) \\ &= 2t \frac{1}{n} + 2\mathbb{E}[X_1 | X_1 < t, \max_{i>1} X_i = t] \frac{n-1}{n} \\ &= 2t \frac{1}{n} + 2 \frac{t}{2} \frac{n-1}{n} \text{ since } X_2, \dots, X_n \text{ are independent of } X_1 \text{ and } X_1 | X_1 < t \sim \text{Uniform}[0, t]. \end{aligned}$$

Hence, $\hat{\theta} = (n+1)/n \max_i X_i$ is an estimator with $\text{mse}(\hat{\theta}) < \text{mse}(\tilde{\theta})$.

Definition 1.7 – Likelihood and maximum likelihood estimator

The likelihood of θ is the function $L : \theta \mapsto f_X(x | \theta)$. The maximum likelihood estimator (MLE) is the value of θ maximising L .

Definition 1.8 – Log-likelihood

The log-likelihood l is $l(\theta) = \log(L(\theta))$.

If $X_1, \dots, X_n \sim f_X(\cdot | \theta)$ are iid, then $L(\theta) = \prod_{i=1}^n f_X(x_i | \theta)$ and $l(\theta) = \sum_{i=1}^n \log(f_X(x_i | \theta))$.

Example 1.8

Let $X_1, \dots, X_n \sim \text{Ber}(p)$ be iid.

$$l(p) = \left(\sum_{i=1}^n x_i \right) \log p + \left(n - \sum_{i=1}^n x_i \right) \log(1 - p)$$

$$\frac{\partial l}{\partial p} = \frac{\sum_i x_i}{p} - \frac{n - \sum_i x_i}{1 - p}$$

so

$$\frac{\partial l}{\partial p} = 0 \iff p = \frac{\sum_i x_i}{n}.$$

Hence, the MLE is $\hat{p} = (1/n) \sum_i x_i$. Note that $\mathbb{E}(\hat{p}) = p$ so $\hat{p}$ is unbiased.

Example 1.9

Suppose $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ are iid.

$$l(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

$$\frac{\partial l}{\partial \mu} = -\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

which is zero when $\mu = (1/n) \sum_i x_i = \bar{X}$ regardless of σ .

$$\frac{\partial l}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2.$$

If we set $\mu = (1/n) \sum_i x_i$, then this is zero when $\sigma^2 = (1/n) \sum_i (x_i - \bar{x})^2 = S_{XX}/n$. Hence, the MLE is $(\hat{\mu}, \hat{\sigma}^2) = (\bar{X}, S_{XX}/n)$. Note that $\hat{\mu}$ is unbiased since $\mathbb{E}(\hat{\mu}) = \mu$. However, we will see later that $S_{XX}/\sigma^2 = n\hat{\sigma}^2/\sigma^2 \sim \chi_{n-1}^2$ so

$$\mathbb{E}(\hat{\sigma}^2) = \mathbb{E}\left(\frac{S_{XX}}{n}\right) = \frac{\sigma^2}{n} \mathbb{E}(\chi_{n-1}^2) = \frac{n-1}{n} \sigma^2$$

and $\hat{\sigma}^2$ is not unbiased but is asymptotically unbiased.

Example 1.10

Suppose $X_1, \dots, X_n \sim \text{Uniform}[0, \theta]$ are iid. Then,

$$L(\theta) = (1/\theta^n) \mathbb{1}(\max_i x_i \leq \theta)$$

so the MLE is $\hat{\theta}_{\text{MLE}} = \max_i X_i$. Recall that in Example 1.7, we had an unbiased estimator $\tilde{\theta}$ and by the Rao-Blackwell theorem, we found $\hat{\theta} = \mathbb{E}(\tilde{\theta} | T) = (n+1)/n \max_i X_i$. Note that $\hat{\theta}_{\text{MLE}} = ((n+1)/n)\hat{\theta}$ so $\mathbb{E}(\hat{\theta}_{\text{MLE}}) = ((n+1)/n)\mathbb{E}(\hat{\theta}) = ((n+1)/n)\theta$ and $\hat{\theta}_{\text{MLE}}$ is asymptotically unbiased.

Some properties of the MLE.

- (i) If T is a sufficient statistic, then the MLE is a function of T . We can factorise $L(\theta) = g(T(x), \theta)h(x)$.
- (ii) If $\phi = h(\theta)$ where h is a bijection, then the MLE of ϕ is $\hat{\phi} = h(\hat{\theta})$ where $\hat{\theta}$ is the MLE of θ .
- (iii) We have asymptotic normality. $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta)$ is approximately normal with mean zero for large n . The covariance matrix is the smallest attainable. A more rigorous statement is given in Part II Principles of Statistics.

Example 1.11 – Stein’s paradox

Suppose we have independent observations $X_i \sim \mathcal{N}(\mu_i, 1)$ and we want to estimate $\mu = (\mu_1, \dots, \mu_p) \in \mathbb{R}^p$. For example, μ_1 could be the mean IQ of a Cambridge student, μ_2 could be the mean radius of a crater on the moon, and μ_3 could be the mean weight of a sheep from New Zealand. Intuition tells us that the best estimator should be $\hat{\mu} = X$. However, there exists unintuitive estimators with smaller mse. The James-Stein estimator is

$$\hat{\mu}_{\text{JS}} = \left(1 - \frac{p-2}{\|X\|_2^2}\right) X.$$

We have

$$\text{mse}(\hat{\mu}_{\text{JS}}) = \mathbb{E}_{\mu}[\|\hat{\mu}_{\text{JS}} - \mu\|_2^2] \leq \mathbb{E}_{\mu}[(\hat{\mu} - \mu)^2] = \text{mse}(\hat{\mu})$$

for all $\mu \in \mathbb{R}^p$. How could this possibly be true? The intuition comes from the bias-variance decomposition. Define $\hat{\mu}_{\lambda} = \lambda X$,

$$\text{mse}(\hat{\mu}_{\lambda}) = \text{bias}(\hat{\mu}_{\lambda})^2 + \text{var}(\hat{\mu}_{\lambda}) = (\lambda - 1)^2 \|\mu\|_2^2 + \lambda^2 p.$$

As a function of λ , we see that for larger p , it is advantageous to choose smaller λ to trade off bias versus variance. The optimal λ depends on $\|\mu\|$, but since we do not know this, $\|X\|_2$ is a good surrogate.

Definition 1.9 – Confidence interval

A $(100\gamma)\%$ confidence interval for a parameter θ is a random interval $(A(X), B(X))$ such that

$$\mathbb{P}(A(X) \leq \theta \leq B(X)) = \gamma$$

for some $\gamma \in (0, 1)$ and all values of the true parameter θ .

Example 1.12

Suppose $X_1, \dots, X_n \sim \mathcal{N}(\theta, 1)$ are iid. We want to find a 95% confidence interval for θ . We know that $\bar{X} = (1/n) \sum_{i=1}^n X_i \sim \mathcal{N}(\theta, 1/n)$. If we define $Z = \sqrt{n}(\bar{X} - \theta)$, then $Z \sim \mathcal{N}(0, 1)$ no matter the value of θ . Let z_1 and z_2 be numbers such that $\Phi(z_2) - \Phi(z_1) = 0.95$ where Φ is the cdf of $\mathcal{N}(0, 1)$.

$$\begin{aligned} \mathbb{P}(z_1 \leq \sqrt{n}(\bar{X} - \theta) \leq z_2) &= 0.95 \\ \mathbb{P}\left(\bar{X} - \frac{z_2}{\sqrt{n}} \leq \theta \leq \bar{X} - \frac{z_1}{\sqrt{n}}\right) &= 0.95 \end{aligned}$$

Hence, $(\bar{X} - z_2/\sqrt{n}, \bar{X} - z_1/\sqrt{n})$ is a 95% confidence interval. A good choice would be $\bar{X} \pm 1.96/\sqrt{n}$.

We can use the following steps to construct a confidence interval.

- (i) Find a quantity $R(X, \theta)$ such that the $\mathbb{P}_{\theta}$ distribution of $R(X, \theta)$ does not depend on θ . This is called a

pivot. In Example 1.12, $R(X, \theta) = \sqrt{n}(\bar{X} - \theta)$.

- (ii) Write down the statement $\mathbb{P}(c_1 \leq R(X, \theta) \leq c_2) = \gamma$ where c_1 and c_2 are the quantiles of the distribution of $R(X, \theta)$.
- (iii) Rearrange the above to leave θ in the middle of the inequality.

When θ is a vector, we talk about confidence sets rather than confidence intervals.

Example 1.13

Suppose $X_1, \dots, X_n \sim \mathcal{N}(0, \sigma^2)$ are iid. We want a 95% confidence interval for σ^2 . Note that $X_i/\sigma \sim \mathcal{N}(0, 1)$ so $\sum_{i=1}^n X_i^2/\sigma^2 \sim \chi_n^2$. Hence, $R(X, \sigma^2) = \sum_{i=1}^n X_i^2/\sigma^2$ is a pivot. Let $c_1 = F_{\chi_n^2}^{-1}(0.025)$ and $c_2 = F_{\chi_n^2}^{-1}(0.975)$ where $F_{\chi_n^2}$ is the cdf of χ_n^2 . Then,

$$\mathbb{P}\left(c_1 \leq \sum_{i=1}^n \frac{X_i^2}{\sigma^2} \leq c_2\right) = 0.95$$

$$\mathbb{P}\left(\frac{\sum_{i=1}^n X_i^2}{c_2} \leq \sigma^2 \leq \frac{\sum_{i=1}^n X_i^2}{c_1}\right) = 0.95.$$

Hence, $(\sum X_i^2/c_2, \sum X_i^2/c_1)$ is a 95% confidence interval for σ^2 . Also, $((\sum X_i^2/c_2)^{1/2}, (\sum X_i^2/c_1)^{1/2})$ is a 95% confidence interval for σ .

Example 1.14

Suppose $X_1, \dots, X_n \sim \text{Ber}(p)$ are iid. Suppose n is large. We will find an approximation of a 95% confidence interval for p . The MLE of p is $\hat{p} = (1/n) \sum_{i=1}^n X_i$. By the central limit theorem, $\hat{p} \approx \mathcal{N}(p, p(1-p)/n)$. Thus,

$$\frac{\sqrt{n}(\hat{p} - p)}{\sqrt{p(1-p)}} \approx \mathcal{N}(0, 1).$$

We have

$$\mathbb{P}\left(-z_{0.025} \leq \frac{\sqrt{n}(\hat{p} - p)}{\sqrt{p(1-p)}} \leq z_{0.025}\right) \approx 0.95.$$

Instead of directly inverting to obtain the confidence interval, we say that if n is large, $\hat{p} \approx p$ so

$$\mathbb{P}\left(\hat{p} - z_{0.025} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{0.025} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) \approx 0.95.$$

We arrive at the confidence interval $\hat{p} \pm z_{0.025} \sqrt{p(1-p)/n}$. Since we have $\hat{p}(1-\hat{p}) \leq 1/4$ for all $\hat{p} \in [0, 1]$, we could also report a conservative confidence interval $\hat{p} \pm z_{0.025} \sqrt{1/(4n)}$.

2 Bayesian estimation

So far, frequentist methods have treated $\theta \in \Theta$ as a fixed unknown. In contrast, Bayesian methods treat θ as random, with a prior distribution $\pi(\theta)$. Conditional on θ , the data X has pdf (or pmf) $f_X(\cdot | \theta)$. Having observed $X = x$, we could combine this with the prior distribution to form the posterior distribution $\pi(\theta | x)$.

By Bayes' rule,

$$\pi(\theta | x) = \frac{\pi(\theta)f_X(x | \theta)}{f_X(x)}$$

where $f_X(x)$ is the marginal distribution of X so

$$f_X(x) = \begin{cases} \int_{\Theta} f_X(x|\theta)\pi(\theta) d\theta & \text{if } \theta \text{ is continuous} \\ \sum_{\Theta} f_X(x|\theta)\pi(\theta) & \text{if } \theta \text{ is discrete.} \end{cases}$$

More simply,

$$\underbrace{\pi(\theta|x)}_{\text{posterior}} \propto \underbrace{\pi(\theta)}_{\text{prior}} \underbrace{f_X(x|\theta)}_{\text{likelihood}}.$$

Often, it is easier to recognise the right hand side as proportional to a known distribution. By the factorisation criterion, the posterior only depends on X through a sufficient statistic.

$$\pi(\theta|x) \propto \pi(\theta)f_X(x|\theta) = \pi(\theta)g(T(x),\theta)h(x) \propto \pi(\theta)g(T(x),\theta).$$

Example 2.1

Suppose $\theta \in [0,1]$ is the mortality rate for a procedure at Addenbrooke's. In the first 10 operations, there are no deaths. In other hospitals around the country, the mortality rate is between 3% and 20%, with an average of 10%. Consider the prior $\pi(\theta) \sim \text{Beta}(a,b)$. We can choose $a = 3$ and $b = 27$ so that $\pi(\theta)$ has mean 0.1 and $\pi(0.03 \leq \theta \leq 0.2) = 0.9$. Let $X_i \sim \text{Ber}(p)$ be an indicator for whether the i -th patient at Addenbrooke's dies. Then, $f_X(x|\theta) = \theta^{\sum x_i} (1-\theta)^{n-\sum x_i}$. The posterior is

$$\begin{aligned} \pi(\theta|x) &\propto \pi(\theta)f_X(x|\theta) \propto \theta^{a-1}(1-\theta)^{b-1}\theta^{\sum x_i}(1-\theta)^{n-\sum x_i} \\ &= \theta^{\sum x_i+a-1}(1-\theta)^{b+n-\sum x_i-1} \\ &\sim \text{Beta}\left(a + \sum x_i, b + n - \sum x_i\right). \end{aligned}$$

Plugging in $n = 10$ and $\sum x_i = 0$, the posterior is $\text{Beta}(3,37)$.

In Example 2.1, the prior and posterior were from the same family of distributions. This is called conjugacy.

Example 2.2

Suppose we put a $\text{Beta}(a,b)$ prior on the parameter θ of kidney cancer death rates in each county in the US. We can estimate $a = 27$ and $b = 58000$ since $a/(a+b) \approx 4.65 \times 10^{-4}$ is the average kidney cancer death rate in the US. Example 2.1 shows that if we observe $\sum_{i=1}^n X_i$ deaths in a county, the posterior mean estimate is

$$\frac{a + \sum x_i}{a + b + n} = \frac{n}{a + b + n} \underbrace{\frac{\sum x_i}{n}}_{\text{county mean}} + \frac{a + b}{a + b + n} \underbrace{\frac{a}{a + b}}_{\text{US mean}}.$$

For large n , we use $\approx \sum x_i/n$ as our estimate. For small n , we use $\approx a/(a+b)$ as our estimate. In between, we shrink our estimate towards $a/(a+b)$. This helps better reflect the death rate in smaller counties.

One of the uses of the posterior is in decision theory.

- (i) We must pick a decision $\delta \in D$.
- (ii) We have a loss function $L(\theta, \delta)$ which gives the loss occurred in making decision δ when the true parameter value is θ .
- (iii) The Von Neumann-Morgenstern theorem states that under axioms of rational behaviour, we should pick δ which minimises expected loss under posterior.

Definition 2.1 – Bayes estimator

Let $h(\delta) = \int_{\Theta} L(\theta, \delta) \pi(\theta | x) d\theta$. The Bayes estimator $\hat{\theta}^{(b)}$ is defined by

$$\hat{\theta}^{(b)} = \operatorname{argmin}_{\delta} h(\delta).$$

Example 2.3

Consider a quadratic loss $L(\theta, \delta) = (\theta - \delta)^2$. Then, $h(\delta) = \int_{\Theta} (\theta - \delta)^2 \pi(\theta | x) d\theta$. Differentiating with respect to δ , we have $h'(\delta) = 0$ if $\int_{\Theta} (\theta - \delta) \pi(\theta | x) d\theta = 0$. Hence,

$$\hat{\theta}^{(b)} = \int_{\Theta} \theta \pi(\theta | x) d\theta$$

is the posterior mean.

Now consider an absolute loss $L(\theta, \delta) = |\theta - \delta|$. Then,

$$\begin{aligned} h(\delta) &= \int_{\Theta} |\theta - \delta| \pi(\theta | x) d\theta \\ &= \int_{-\infty}^{\delta} -(\theta - \delta) \pi(\theta | x) d\theta + \int_{\delta}^{\infty} (\theta - \delta) \pi(\theta | x) d\theta \\ &= -\int_{-\infty}^{\delta} \theta \pi(\theta | x) d\theta + \int_{\delta}^{\infty} \theta \pi(\theta | x) d\theta + \delta \int_{-\infty}^{\delta} \pi(\theta | x) d\theta - \delta \int_{\delta}^{\infty} \pi(\theta | x) d\theta \\ h'(\delta) &= \int_{-\infty}^{\delta} \pi(\theta | x) d\theta - \int_{\delta}^{\infty} \pi(\theta | x) d\theta. \end{aligned}$$

Hence, $h'(\delta) = 0$ if and only if

$$\int_{-\infty}^{\delta} \pi(\theta | x) d\theta = \int_{\delta}^{\infty} \pi(\theta | x) d\theta,$$

so $\hat{\theta}^{(b)}$ is the posterior median.

Example 2.4

Suppose $X_1, \dots, X_n \sim \mathcal{N}(\mu, 1)$ are iid with prior $\pi(\mu)$ that is $\mathcal{N}(0, 1/\tau^2)$ for some known $\tau > 0$. Then,

$$\begin{aligned} \pi(\mu | x) &\propto f_X(x | \mu) \pi(\mu) \propto \exp \left[-\frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2 \right] \exp \left[-\frac{\mu^2 \tau^2}{2} \right] \\ &\propto \exp \left[-\frac{1}{2} (n + \tau^2) \left(\mu - \frac{\sum x_i}{n + \tau^2} \right)^2 \right] \\ &\sim \mathcal{N} \left(\frac{\sum x_i}{n + \tau^2}, \frac{1}{n + \tau^2} \right). \end{aligned}$$

The posterior mean and median are both $\sum x_i / (n + \tau^2)$. Comparing with the MLE, we observe that

$$\hat{\theta}^{(b)} = \frac{n}{n + \tau^2} \underbrace{\frac{\sum x_i}{n}}_{\hat{\theta}_{\text{MLE}}} + \frac{\tau^2}{n + \tau^2} (0).$$

Definition 2.2 – Credible interval

A $(100\gamma)\%$ credible interval $(A(X), B(X))$ satisfies

$$\pi(A(X) \leq \theta \leq B(X) | X = x) = \gamma.$$

Example 2.5

There is a jar of coins. If you could guess the exact number of coins in said jar, you get all the coins. We suppose a prior $\mathcal{N}(\mu, \sigma^2)$. If we guessed x coins, the expected earning is

$$f(x) = x \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Maximising $f(x)$, the number of coins we should guess is $x = (\mu + \sqrt{\mu^2 + 4\sigma^2})/2$, instead of the intuitive $x = \mu$.

3 Hypothesis testing

A hypothesis is an assumption about the distribution of data X taking values in $\mathcal{X}$. A null hypothesis H_0 describes a base case with no effect. An alternative hypothesis H_1 describes a more interesting cases with a positive or negative effect.

Example 3.1

Suppose $X_1, \dots, X_n \sim \text{Ber}(\theta)$ are iid. Our null hypothesis could describe a fair coin so $H_0 : \theta = 1/2$. Our alternative hypothesis might be $H_1 : \theta \neq 1/2$.

Example 3.2

Suppose $X_1, \dots, X_n$ are iid. The goodness of fit test considers $H_0 : X_i$ has pdf f_0 and $H_1 : X_i$ has pdf f_1 .

Example 3.3

Suppose X has pdf $f(\cdot | \theta)$ for some $\theta \in \Theta$. We could have $H_0 : \theta \in \Theta_0 \subsetneq \Theta$ and $H_1 : \theta \notin \Theta_0$.

A simple hypothesis fully specifies the distribution of X . Otherwise, it is a composite hypothesis. A test of H_0 is defined by a critical region C . When $X \in C$, we reject H_0 . When $X \notin C$, we do not reject H_0 .

There are two types of error. A type I error is when we reject H_0 when H_0 is true. A type II error is when we fail to reject H_0 when H_1 is true.

When H_0 and H_1 are simple hypothesis, we write

$$\alpha = \mathbb{P}_{H_0}(H_0 \text{ is rejected}) = \mathbb{P}_{H_0}(X \in C)$$

as the size of the test, which is the probability of a type I error. Similarly, we write

$$\beta = \mathbb{P}_{H_1}(H_0 \text{ is not rejected}) = \mathbb{P}_{H_1}(X \notin C)$$

as the probability of a type II error. We call $1 - \beta$ the power of the test. Usually, we set α at an acceptable level, for example 1%, and choose a test that minimises β , subject to $\alpha \leq 1\%$.

Lemma 3.1 – Neyman-Pearson lemma

Let H_0 and H_1 be simple hypothesis, with X having pdf f_i under H_i . The likelihood ratio statistic is

$$\Lambda_X(H_0, H_1) = \frac{f_1(X)}{f_0(X)}.$$

A likelihood ratio test (LRT) rejects H_0 when $X \in C = \{x : \Lambda_x(H_0, H_1) > k\}$ for some $k > 0$.

Theorem 3.1

Suppose f_0 and f_1 are non-zero on the same sets, and there exists k such that the LRT with critical region $C = \{x : f_1(x)/f_0(x) > k\}$ has size α . Out of all the tests with size $\leq \alpha$, the LRT is the test with the smallest β .

Proof. Let $\bar{C}$ be the complement of C .

$$\alpha = \mathbb{P}_{H_0}(X \in C) = \int_C f_0(x) dx \quad \beta = \mathbb{P}_{H_1}(X \in \bar{C}) = \int_{\bar{C}} f_1(x) dx$$

Let C^* be the critical region of another test of size $\alpha^* \leq \alpha$. We want to show $\beta \leq \beta^*$.

$$\begin{aligned} \beta - \beta^* &= \int_{\bar{C}} f_1(x) dx - \int_{\bar{C}^*} f_1(x) dx \\ &= \int_{\bar{C} \cap C^*} f_1(x) dx - \int_{\bar{C}^* \cap C} f_1(x) dx \\ &= \int_{\bar{C} \cap C^*} \underbrace{\frac{f_1(x)}{f_0(x)}}_{\leq k \text{ in } \bar{C}} f_0(x) dx - \int_{\bar{C}^* \cap C} \underbrace{\frac{f_1(x)}{f_0(x)}}_{> k \text{ in } C} f_0(x) dx \\ &\leq k \left[\int_{\bar{C} \cap C^*} f_0(x) dx - \int_{\bar{C}^* \cap C} f_0(x) dx \right] \\ &= k \left[\int_{C^*} f_0(x) dx - \int_C f_0(x) dx \right] \\ &= k(\alpha^* - \alpha) \leq 0 \end{aligned}$$

□

Note that a LRT with size α does not always exist for any given α . However, we can always define a randomised test with exact size α .

Example 3.4

Suppose $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma_0^2)$ are iid where σ_0^2 is known. We want to find the best size α test for $H_0 : \mu = \mu_0$ versus $H_1 : \mu = \mu_1$ for some fixed $\mu_1 > \mu_0$.

$$\Lambda_x(H_0, H_1) = \frac{(2\pi\sigma_0)^{-n/2} \exp \left[-\frac{1}{2\sigma_0^2} \sum_{i=1}^n (x_i - \mu_1)^2 \right]}{(2\pi\sigma_0)^{-n/2} \exp \left[-\frac{1}{2\sigma_0^2} \sum_{i=1}^n (x_i - \mu_0)^2 \right]} = \exp \left[\frac{\mu_1 - \mu_0}{\sigma_0^2} (n\bar{x}) + \frac{n(\mu_0^2 - \mu_1^2)}{2\sigma_0^2} \right].$$

Since $\Lambda_x(H_0, H_1)$ is monotone in $\bar{x}$, we have $\Lambda_x(H_0, H_1) > k$ if and only if $\bar{x} > c$ for some c . By the same logic, if $z = \sqrt{n}(\bar{x} - \mu_0)/\sigma_0$, the rejection region is $\{z > c'\}$ for some c' . Under H_0 , $Z \sim \mathcal{N}(0, 1)$ so the test that rejects H_0 when $\{x : z > \Phi^{-1}(1 - \alpha)\}$ has size α . This is called a z-test.

Definition 3.1 – p-value

For any test with critical region of the form $\{x : T(x) > k\}$ where T is some statistic, we usually report the p-value

$$p = \mathbb{P}_{H_0}(T(X) > T(x^*))$$

where x^* is the observed data. In other words, this is the probability of observing more extreme data under H_0 .

Proposition 3.1

Under H_0 , the p -value is Uniform $[0, 1]$.

Proof. Let F be the distribution function of T . Then,

$$\begin{aligned}\mathbb{P}_{H_0}(p < u) &= \mathbb{P}_{H_0}(1 - F(T) < u) \\ &= \mathbb{P}_{H_0}(F(T) > 1 - u) \\ &= \mathbb{P}_{H_0}(T > F^{-1}(1 - u)) \\ &= 1 - F(F^{-1}(1 - u)) = u\end{aligned}$$

where T represents $T(X)$ when $X \sim H_0$. □

Theorem 3.2 – Duality between tests and confidence intervals

Define the acceptance region A of a test as the complement of the critical region. Let $X \sim f_X(\cdot | \theta)$ for some $\theta \in \Theta$.

- (i) Suppose that for each $\theta_0 \in \Theta$, there exists a test of $H_0 : \theta = \theta_0$ of size α with acceptance region $A(\theta_0)$. Then, the set $I(X) = \{\theta : X \in A(\theta)\}$ is a $100(1 - \alpha)\%$ confidence set.
- (ii) Suppose $I(X)$ is a $100(1 - \alpha)\%$ confidence set for θ . Then, $A(\theta_0) = \{X : \theta_0 \in I(X)\}$ is the acceptance region of a size α test for $H_0 : \theta = \theta_0$ for each $\theta_0 \in \Theta$.

Proof. In both cases, $\theta_0 \in I(X)$ if and only if $X \in A(\theta_0)$. For (i), we want to show that $\mathbb{P}_{\theta_0}(I(X) \ni \theta_0) = 1 - \alpha$. The left hand side is $\mathbb{P}_{\theta_0}(X \in A(\theta_0)) = \mathbb{P}_{\theta_0}(\text{do not reject } H_0) = 1 - \alpha$. For (ii), we want to show that $\mathbb{P}_{\theta_0}(X \notin A(\theta_0)) = \alpha$. The left hand side is $\mathbb{P}_{\theta_0}(I(X) \not\ni \theta_0) = \alpha$. □

Example 3.5

Suppose $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma_0^2)$ are iid where σ_0^2 is known. We found a $100(1 - \alpha)\%$ confidence interval for μ in Example 1.12, namely, $I(X) = \bar{X} \pm (\sigma_0/\sqrt{n})z_{\alpha/2}$. Using part (ii) of Theorem 3.2, we can find a size α test for $H_0 : \mu = \mu_0$ by defining the acceptance region

$$A(\mu_0) = \{x : I(x) \ni \mu_0\} = \left\{x : \mu_0 \in \left[\bar{x} - \frac{z_{\alpha/2}\sigma_0}{\sqrt{n}}, \bar{x} + \frac{z_{\alpha/2}\sigma_0}{\sqrt{n}}\right]\right\}.$$

Equivalently, reject H_0 when $\sqrt{n}|\mu_0 - \bar{x}|/\sigma_0 > z_{\alpha/2}$. This is a two-sided LRT. We could also go in the opposite direction and turn a LRT into a confidence interval.

Previously, we considered H_0 and H_1 to be simple hypothesis, with error probabilities $\alpha = \mathbb{P}_{H_0}(X \in C)$ and $\beta = \mathbb{P}_{H_1}(X \notin C)$. Now, we consider composite hypothesis, with $H_0 : \theta \in \Theta_0 \subseteq \Theta$ and $H_1 : \theta \in \Theta_1 \subseteq \Theta$.

Definition 3.2 – Uniformly most powerful test

The power function is $W(\theta) = \mathbb{P}_{\theta}(X \in C)$. The size of a test with composite null H_0 is the worst-case type I error probability $\alpha = \sup_{\theta \in \Theta_0} W(\theta)$. We say that a test of H_0 against H_1 is uniformly most powerful (UMP) of size α if

- (i) $\sup_{\theta \in \Theta_0} W(\theta) \leq \alpha$;
- (ii) for any other test of size $\leq \alpha$ with power function W^* , we have $W(\theta) \geq W^*(\theta)$ for all $\theta \in \Theta_1$.

Note that UMP tests might not exist. However, many LRTs are UMP.

Example 3.6 – One-sided test for normal location

Suppose $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma_0^2)$ are iid where σ_0^2 is known. We wish to test $H_0 : \mu \leq \mu_0$ versus $H_1 : \mu > \mu_0$. Recall that for the simple hypothesis $H'_0 : \mu = \mu_0$ versus $H'_1 : \mu = \mu_1$ with $\mu_0 < \mu_1$, the LRT had critical region $C = \{x : z = \sqrt{n}(\bar{x} - \mu_0)/\sigma_0 > z_\alpha\}$. We will show that the same test is UMP for H_0 against H_1 . The power function is

$$\begin{aligned} W(\mu) &= \mathbb{P}_\mu(\text{reject } H_0) \\ &= \mathbb{P}_\mu \left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} > z_\alpha \right) \\ &= \mathbb{P}_\mu \left(\frac{\sqrt{n}(\bar{x} - \mu)}{\sigma_0} > z_\alpha + \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma_0} \right) \\ &= 1 - \Phi \left(z_\alpha + \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma_0} \right) \end{aligned}$$

since $\sqrt{n}(\bar{x} - \mu)/\sigma_0 \sim \mathcal{N}(0, 1)$ under $\mathbb{P}_\mu$. Thus, $\sup_{\mu \in \Theta_0} W(\mu) = \alpha$ so the test has size α . Now consider any other test of size $\leq \alpha$ with power function W^* . We want to show that $W(\mu_1) \geq W^*(\mu_1)$ for all $\mu_1 \in \Theta_1$. Any other test of size $\leq \alpha$ also has size $\leq \alpha$ for H'_0 against H'_1 since $W^*(\mu_0) \leq \sup_{\mu \in \Theta_0} W^*(\mu) \leq \alpha$. By the Neyman-Pearson Lemma, $W(\mu_1) \geq W^*(\mu_1)$. Since this argument works for any $\mu_1 > \mu_0$, we are done.

Definition 3.3 – Generalised likelihood ratio statistic

The generalised likelihood ratio (GLR) statistic for $H_0 : \theta \in \Theta_0 \subseteq \Theta$ versus $H_1 : \theta \in \Theta_1 \subseteq \Theta$ is

$$\Lambda_X(H_0, H_1) = \frac{\sup_{\theta \in \Theta_1} f_X(\cdot | \theta)}{\sup_{\theta \in \Theta_0} f_X(\cdot | \theta)}.$$

We reject H_0 when Λ_X is large.

Example 3.7 – Two-sided test for normal location

Suppose $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma_0^2)$ are iid where σ_0^2 is known. We wish to test $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$. Thus, $\Theta_0 = \{\mu_0\}$ and $\Theta_1 = \mathbb{R} \setminus \{\mu_0\}$. The GLR statistic is

$$\Lambda_x(H_0, H_1) = \frac{(2\pi\sigma_0^2)^{-n/2} \exp \left[-\frac{1}{2\sigma_0^2} \sum_{i=1}^n (x_i - \bar{x})^2 \right]}{(2\pi\sigma_0^2)^{-n/2} \exp \left[-\frac{1}{2\sigma_0^2} \sum_{i=1}^n (x_i - \mu_0)^2 \right]}.$$

Note that

$$2 \log \Lambda_x(H_0, H_1) = \frac{1}{\sigma_0^2} \left(\sum (x_i - \mu_0)^2 - \sum (x_i - \bar{x})^2 \right) = \frac{n}{\sigma_0^2} (\bar{x} - \mu_0)^2.$$

Thus, the GLRT rejects H_0 if $\sqrt{n}|\bar{x} - \mu_0|/\sigma_0$ is large. Under H_0 , $\sqrt{n}(\bar{x} - \mu_0)/\sigma_0 \sim \mathcal{N}(0, 1)$, so a test of size α rejects H_0 if

$$\frac{\sqrt{n}|\bar{x} - \mu_0|}{\sigma_0} > z_{\alpha/2}.$$

Note that $2 \log \Lambda_X(H_0, H_1) \sim \chi_1^2$. Thus, the critical region of the GLRT can also be written as

$$\left\{ x : \frac{n(\bar{x} - \mu_0)^2}{\sigma_0^2} > \chi_1^2(\alpha) \right\}.$$

In fact, a more general result says that $2 \log \Lambda_X(H_0, H_1) \approx \chi^2$ distribution when n is large.

Definition 3.4 – Dimension

Suppose the parameter $\theta = (\theta_1, \dots, \theta_k)$ is k -dimensional. The dimension of a hypothesis $H_0 : \theta \in \Theta_0$ is the number of free parameters in Θ_0 .

Example 3.8

- (i) If $\Theta_0 = \{\theta \in \mathbb{R}^k : \theta_1 = \theta_2 = \dots = \theta_p = 0\}$, then $\dim(\Theta_0) = k - p$.
- (ii) If $\Theta_0 = \{\theta \in \mathbb{R}^k : \theta_i = f_i(\phi) \text{ for all } 1 \leq i \leq k \text{ for some } \phi \in \mathbb{R}^{k-p}\}$ where the f_i 's are not necessarily linear, then $\dim(\Theta_0) = k - p$.

Theorem 3.3 – Wilk's theorem

Suppose $\Theta_0 \subseteq \Theta_1$ and $\dim(\Theta_1) - \dim(\Theta_0) = p$. If $X = (X_1, \dots, X_n)$ are iid, then under regularity conditions, as $n \rightarrow \infty$, the limiting distribution of $2 \log \Lambda_X(H_0, H_1)$ under H_0 is χ_p^2 . Thus, if we reject H_0 when $2 \log \Lambda_X(H_0, H_1) \geq \chi_p^2(\alpha)$, we have a test of size $\approx \alpha$.

Proof. A sketch only.

$$\log \Lambda_X(H_0, H_1) = \sum_{i=1}^n \log \left(\frac{f_X(x_i | \hat{\theta}_0)}{f_X(x_i | \hat{\theta}_1)} \right)$$

and we Taylor expand. □

Example 3.9

In Example 3.7, we had $\Theta_0 = \{\mu_0\}$ and $\Theta_1 = \mathbb{R} \setminus \{\mu_0\}$. In this case, $\dim(\Theta_1) - \dim(\Theta_0) = 1$ and we saw that $2 \log \Lambda_X(H_0, H_1) \sim \chi_1^2$ exactly. In a different parametric family, this would hold approximately for large n .

Example 3.10 – Goodness-of-fit testing

In one of his experiments, Mendel crossed 556 smooth yellow male peas with wrinkled green female peas. Here is what he obtained and comparison with predictions from genetic theory.

Type	Observed count	Predicted frequency
Smooth yellow	315	9/16
Smooth green	108	3/16
Wrinkled yellow	102	3/16
Wrinkled green	31	1/16

Is there evidence in the data to reject the hypothesis that the genetic theory is correct? Suppose $X_1, \dots, X_n$ are iid samples from a distribution on $\{1, \dots, k\}$. Let $p_i = \mathbb{P}(X_1 = i)$ and N_i be the number of observations in $\{X_1, \dots, X_n\}$ of type i . Then, $(N_1, \dots, N_k) \sim \text{Multi}(n, p_1, \dots, p_k)$. The likelihood is $L(p) \propto p_1^{N_1} \dots p_k^{N_k}$ so $l(p) = \text{const.} + \sum_{i=1}^k N_i \log(p_i)$. We can test H_0 versus H_1 using a GLRT.

$$2 \log \Lambda = 2 \left(\sup_{p \in \Theta_1} l(p) - \underbrace{\sup_{p \in \Theta_0} l(p)}_{\text{set to be } l(\bar{p})} \right).$$

Note that

$$\sup_{p \in \Theta_1} = \sup_{p : \sum p_i = 1} \sum_{i=1}^k N_i \log(p_i).$$

Using the Lagrangian $\mathcal{L}(p, \lambda) = \sum N_i \log(p_i) - \lambda(\sum p_i - 1)$, we can check that the MLE $\hat{p}$ is given by $\hat{p}_i = N_i/n$. So,

$$2 \log \Lambda = 2 \sum_{i=1}^k N_i \log \left(\frac{N_i}{n \hat{p}_i} \right).$$

Wilk's theorem says $2 \log \Lambda \approx \chi_p^2$ with $p = \dim(\Theta_1) - \dim(\Theta_0) = (k-1) - 0 = k-1$. Thus, we reject H_0 if $2 \log \Lambda \geq \chi_{k-1}^2(\alpha)$. It is common to write

$$2 \log \Lambda = 2 \sum_{i=1}^k o_i \log \left(\frac{o_i}{e_i} \right)$$

where o_i is the observed number of type i and e_i is the expected number of type i . Let $\delta_i = o_i - e_i$. Then,

$$2 \log \Lambda = 2 \sum (e_i + \delta_i) \log \left(1 + \frac{\delta_i}{e_i} \right) = 2 \sum (e_i + \delta_i) \left(\frac{\delta_i}{e_i} - \frac{\delta_i^2}{2e_i^2} + \dots \right) \approx \sum \frac{\delta_i^2}{e_i}.$$

This gives

Definition 3.5 – Pearson's chi-square statistic

$$\sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}$$

We return to the pea plants. Let (p_1, p_2, p_3, p_4) be the proportions of each type. We test $H_0 : p_i = \tilde{p}_i$ for all $1 \leq i \leq 4$ versus $H_1 : \sum p_i = 1$, where $\tilde{p} = (9/16, 3/16, 3/16, 1/16)$. Plugging in the values, $\sum_i (o_i - e_i)^2 / e_i \approx 0.604$ and $\chi_3^2(0.05) \approx 7.815 > 0.604$ so we do not reject H_0 . The p -value is 0.96. A lot of Mendel's experiments had a high p -value. However, recall that the distribution of the p -value is $\text{Uniform}[0, 1]$ under H_0 . This suggests that he repeated his experiments if he got a small p -value.

Suppose now we wish to test $H_0 : p_i = p_i(\theta)$ for some θ against $H_1 : \sum p_i = 1$.

Example 3.11

Consider $H_0 : p_1 = \theta^2, p_2 = 2\theta(1 - \theta), p_3 = (1 - \theta)^2$ where $\theta \in [0, 1]$. The GLRT gives

$$\begin{aligned} 2 \log \Lambda &= 2 \left(\sup_{p: \sum p_i = 1} l(p) - \sup_{\theta} l(p(\theta)) \right) \\ &= 2(l(\hat{p}) - l(p(\hat{\theta}))) \\ &= 2 \sum N_i \log \left(\frac{N_i}{n p_i(\hat{\theta})} \right) \\ &= 2 \sum o_i \log \left(\frac{o_i}{e_i} \right) \approx \sum_i \frac{(o_i - e_i)^2}{e_i}. \end{aligned}$$

One can perform a test with χ_p^2 where $p = \dim(\Theta_1) - \dim(\Theta_0) = 2 - 1 = 1$.

Suppose $(X_1, Y_1), \dots, (X_n, Y_n)$ are iid with X_i taking values in $\{1, \dots, r\}$ and Y_i taking values in $\{1, \dots, c\}$. We wish to test the null hypothesis that X_i and Y_i are independent.

Definition 3.6 – Contingency table

The entries in a contingency table are

$$N_{ij} = |\{l \in \{1, \dots, n\} : (X_l, Y_l) = (i, j)\}|$$

which is the number of samples of type (i, j) .

Example 3.12 – Test for independence

Consider the Part III admissions statistics from a recent year, given by gender and stream.

Course	Female	Male	Other
MASA	10	9	1
MASTH	11	59	0
MASP	10	56	2
MASS	6	23	0
MMATH	13	87	1

We wish to test the null hypothesis that gender and stream are independent. We observed n samples. We construct our probability model by assuming that a sample is of type (i, j) with probability p_{ij} . Then, $(N_{11}, N_{12}, \dots, N_{1c}, N_{21}, \dots, N_{rc}) \sim \text{Multi}(n, p_{11}, p_{12}, \dots, p_{rc})$. Let $p_{i+} = \sum_j p_{ij}$ and $p_{+j} = \sum_i p_{ij}$. Our null hypothesis is $H_0 : p_{ij} = p_{i+}p_{+j}$ for all (i, j) . The alternative hypothesis is $H_1 : \{p_{ij}\}$ unconstrained with $\sum_{i,j} p_{ij} = 1$.

Using Lagrangian methods, the MLEs are $\hat{p}_{ij} = N_{ij}/n$ under H_1 and $(\hat{p}_{i+}, \hat{p}_{+j}) = (N_{i+}/n, N_{+j}/n)$ under H_0 . Thus, the GLR statistic is

$$2 \log \Lambda = 2 \sum_{i=1}^r \sum_{j=1}^c N_{ij} \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+} \hat{p}_{+j}} \right).$$

Writing the observed count $o_{ij} = N_{ij}$ and the expected count $e_{ij} = n \hat{p}_{i+} \hat{p}_{+j}$, we have

$$2 \log \Lambda = 2 \sum_{i=1}^r \sum_{j=1}^c o_{ij} \log \left(\frac{o_{ij}}{e_{ij}} \right) \approx \sum_{i=1}^r \sum_{j=1}^c \frac{(o_{ij} - e_{ij})^2}{e_{ij}}.$$

By Wilk's theorem, this converges to χ_p^2 with degrees of freedom

$$p = \dim(\Theta_1) - \dim(\Theta_0) = (rc - 1) - ((r - 1) + (c - 1)) = (r - 1)(c - 1).$$

Returning to the data, $\sum \sum (o_{ij} - e_{ij})^2 / e_{ij} \approx 22.2$. The p -value is $\mathbb{P}(\chi_8^2 \geq 22.2) \approx 0.005$ so we reject the null hypothesis of independence.

Example 3.13

Consider the MMATH admissions statistics by gender and year.

Year	Female	Male	Other
2021	13	87	1
2022	10	81	0
2023	15	87	4
2024	9	103	0
2025	16	126	0

Our null hypothesis is once again that gender and year are independent. $\sum \sum (o_{ij} - e_{ij})^2 / e_{ij} \approx 15.5$. The

p -value is $P(\chi_8^2 \geq 15.5) \approx 0.051$ so we do not reject H_0 at 5% significance level.

Under a similar setting, we wish to test for homogeneity.

Example 3.14 – Test for homogeneity

Suppose 150 patients are randomly allocated to three groups of equal size. Two sets of patients received a drug of two different doses. The third group received a placebo.

	Improved	No difference	Worsened
Placebo	18	12	15
Treatment A	20	10	20
Treatment B	25	13	12

This is different from the last setting since the row totals are fixed. We want to test the null hypothesis of homogeneity, that is H_0 corresponds to the probability of the response unchanged between groups. Our probability model would be $(N_{i1}, \dots, N_{ic}) \sim \text{Multi}(n_{i+}, p_{i1}, \dots, p_{ic})$ independently for each $i \in \{1, \dots, r\}$, where the n_{i+} 's are the row totals. The null hypothesis is $H_0 : p_{1j} = p_{2j} = \dots = p_{rj}$ for all $j \in \{1, \dots, c\}$. The alternative hypothesis is $H_1 : (p_{i1}, \dots, p_{ic})$ is an arbitrary probability distribution for each $i \in \{1, \dots, r\}$.

Under H_1 ,

$$L(p) = \prod_{i=1}^r \frac{n_{i+}!}{N_{i1}! \dots N_{ic}!} p_{i1}^{N_{i1}} \dots p_{ic}^{N_{ic}}$$

so $l(p) = \text{const.} + \sum_{i,j} N_{ij} \log p_{ij}$. Using the Lagrangian method with the constraints $\sum_j p_{ij} = 1$ for all $i \in \{1, \dots, r\}$, we find that the MLE is $\hat{p}_{ij} = N_{ij}/n_{i+}$. Under H_0 , let $p_{ij} = p_j$ for all j , then

$$l(p) = \text{const.} + \sum_{i,j} N_{ij} \log p_{ij} = \text{const.} + \sum_{j=1}^c N_{+j} \log p_j.$$

Using the Lagrangian method with constraint $\sum_j p_j = 1$, we have the MLE $\hat{p}_j = N_{+j}/n_{++}$. Hence,

$$2 \log \Lambda = 2 \sum_{i=1}^r \sum_{j=1}^c N_{ij} \log \left(\frac{\hat{p}_{ij}}{\hat{p}_j} \right) = 2 \sum_{i=1}^r \sum_{j=1}^c N_{ij} \log \left(\frac{N_{ij}}{n_{i+} N_{+j} / n_{++}} \right).$$

If we define $o_{ij} = N_{ij}$ and $e_{ij} = n_{i+} N_{+j} / n_{++}$, then

$$2 \log \Lambda = 2 \sum_{i,j} o_{ij} \log \left(\frac{o_{ij}}{e_{ij}} \right) \approx \sum_{i,j} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}.$$

Wilk's theorem states that asymptotically, the distribution of $2 \log \Lambda$ is χ_p^2 , with

$$p = \dim(\Theta_1) - \dim(\Theta_0) = r(c-1) - (c-1) = (r-1)(c-1).$$

We see that the degrees of freedom is the same for both tests of independence and homogeneity.

4 Multivariate normal theory

Let $X = (X_1 \dots X_n)^\top$ be a vector of random variables. Recall that

$$\mathbb{E}[X] = (\mathbb{E}(X_1) \dots \mathbb{E}(X_n))^\top \quad \text{Cov}(X) = \mathbb{E}[(X - \mathbb{E}(X))(X - \mathbb{E}(X))^\top] = (\text{cov}(X_i, X_j))_{ij}.$$

Furthermore, if $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, then

$$\mathbb{E}(AX + b) = A\mathbb{E}(X) + b$$

$$\text{Cov}(AX + b) = A \text{Cov}(X) A^\top.$$

Definition 4.1 – Multivariate normal distribution

We say that X has a multivariate normal distribution if for any $t \in \mathbb{R}^n$, the random variable $t^\top X$ has a normal distribution.

Proposition 4.1

If X is multivariate normal, then $AX + b$ is multivariate normal.

Proof. Take any $t \in \mathbb{R}^m$, then $t^\top (AX + b) = (A^\top t)^\top X + t^\top b$. Since X is multivariate normal, $(A^\top t)^\top X \sim \mathcal{N}(\mu, \sigma^2)$ for some (μ, σ^2) . Hence, $t^\top (AX + b) \sim \mathcal{N}(\mu + t^\top b, \sigma^2)$. $\square$

Proposition 4.2

A multivariate normal distribution is fully specified by its mean and covariance.

Proof. Let X_1 and X_2 each be multivariate normal with mean $\mu \in \mathbb{R}^n$ and covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$. We will show that their moment generating functions are equal, so the distributions are equal.

$$M_{X_1}(t) = \mathbb{E}[e^{t^\top X_1}] = \exp\left(\mathbb{E}[t^\top X_1] + \frac{1}{2} \text{var}(t^\top X_1)\right) = \exp\left(t^\top \mu + \frac{1}{2} t^\top \Sigma t\right)$$

since $t^\top X_1$ is normal. As the calculation is the same for X_2 , they have the same distribution. $\square$

Corollary 4.1

Suppose $X_1, \dots, X_n \sim \mathcal{N}(\mu_i, \sigma^2)$ are independent. Let $A \in \mathbb{R}^{n \times n}$ be an orthogonal matrix, so $A^\top A = I$. Then, the elements of $Y = AX \in \mathbb{R}^n$ are independently distributed, and $Y_i \sim \mathcal{N}((A\mu)_i, \sigma^2)$.

Proof. The distribution of Y is determined by its mean vector and its covariance matrix.

$$\mathbb{E}[Y] = \mathbb{E}[AX] = A\mathbb{E}[X] = A\mu$$

$$\text{Cov}(AX) = A \text{Cov}(X) A^\top = A(\sigma^2 I) A^\top = \sigma^2 I$$

$\square$

Proposition 4.3

Let $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ be iid for some unknown $\mu \in \mathbb{R}$ and $\sigma^2 > 0$. Recall that the MLE of μ is $\bar{X} = (1/n) \sum_{i=1}^n X_i$, and the MLE of σ^2 is $\hat{\sigma}^2 = S_{XX}/n = (\sum_{i=1}^n (X_i - \bar{X})^2)/n$. Then,

- (i) $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$;
- (ii) $S_{XX}/\sigma^2 \sim \chi_{n-1}^2$;
- (iii) $\bar{X}$ and S_{XX} are independent.

Proof. (i) is an easy consequence of the fact that linear combinations of Gaussians are Gaussian. For (ii) and (iii), we write

$$\begin{aligned}\sum_{i=1}^n (X_i - \mu)^2 &= \sum_{i=1}^n [(X_i - \bar{X}) + (\bar{X} - \mu)]^2 \\ &= \sum_{i=1}^n [(X_i - \bar{X})^2 + 2(X_i - \bar{X})(\bar{X} - \mu) + (\bar{X} - \mu)^2] \\ &= S_{XX} + n(\bar{X} - \mu)^2.\end{aligned}$$

Let A be an orthogonal matrix such that

$$Y = A \left[\begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix} - \begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix} \right] = \begin{pmatrix} \sqrt{n}(\bar{X} - \mu) \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \quad \text{by taking } A = \begin{pmatrix} 1/\sqrt{n} & \dots & 1/\sqrt{n} \\ * & \dots & * \\ \vdots & & \vdots \\ * & \dots & * \end{pmatrix}$$

where $*$ are chosen to make A orthogonal. Then, $Y_1 = \sqrt{n}(\bar{X} - \mu) \sim \mathcal{N}(0, \sigma^2)$ and Y_1 is independent of $(Y_2, \dots, Y_n)$ by Corollary 4.1. Since $\sum_{i=1}^n Y_i^2 = \sum_{i=1}^n (X_i - \mu)^2$, we must have

$$\sum_{i=2}^n Y_i^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2 = S_{XX}.$$

Thus, S_{XX} is independent of Y_1 and so is independent of $\bar{X}$. Also, $S_{XX}/\sigma^2 = \sum_{i=2}^n Y_i^2/\sigma^2$ has a χ_{n-1}^2 distribution. $\square$

Definition 4.2 – t -distribution

If $U \sim \mathcal{N}(0, 1)$ and $V \sim \chi_n^2$ with U independent of V , then

$$T = \frac{U}{\sqrt{V/n}}$$

has a t -distribution with n degrees of freedom, denoted t_n .

Example 4.1 – t -test

Suppose $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ are iid. When σ^2 is known, we found a $100(1 - \alpha)\%$ confidence interval for μ by noting that $z = \sqrt{n}(\bar{X} - \mu)/\sigma \sim \mathcal{N}(0, 1)$, so $\bar{X} \pm z_{\alpha/2}\sigma/\sqrt{n}$ is a confidence interval for μ . What if σ^2 is unknown? Since $\bar{X}$ and $\hat{\sigma}^2 = S_{XX}/n$ are independent, with $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ and $S_{XX}/\sigma^2 \sim \chi_{n-1}^2$, we have the pivot

$$T = \frac{\sqrt{n}(\bar{X} - \mu)/\sigma}{\sqrt{S_{XX}/(\sigma^2(n-1))}} \sim t_{n-1}.$$

Thus,

$$\mathbb{P}_{\mu, \sigma^2} \left(-t_{n-1}(\alpha/2) \leq \frac{\sqrt{n}(\bar{X} - \mu)}{\sqrt{S_{XX}/(n-1)}} \leq t_{n-1}(\alpha/2) \right) = 1 - \alpha$$

so

$$\bar{X} \pm \sqrt{\frac{S_{XX}}{n-1}} \frac{t_{n-1}(\alpha/2)}{\sqrt{n}}$$

is an exact $100(1 - \alpha)\%$ confidence interval for μ .

If instead we wished to test $H_0 : \mu = \mu_0$ with σ^2 unknown, then under H_0 , we have $T \sim t_{n-1}$. So, we reject H_0 when $|T| \geq t_{n-1}(\alpha/2)$. This is known as a t -test. As $n \rightarrow \infty$, we have $S_{XX}/(n-1) \rightarrow \sigma^2$ and $t_{n-1}(\alpha/2) \rightarrow z_{\alpha/2}$. Then, the t -test converges to a z -test. Importantly, the validity of the t -test does not require large n .

The t -test can be seen as an example of a GLRT for $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$. The GLR statistic is

$$\Lambda_X(H_0, H_1) = \frac{\sup_{\mu, \sigma^2} f(x | \mu, \sigma^2)}{\sup_{\sigma^2} f(x | \mu_0, \sigma^2)}.$$

Plugging in $(\hat{\mu}, \hat{\sigma}^2) = (\bar{X}, S_{XX}/n)$, the numerator is $(2\pi \sum (X_i - \bar{X})^2/n)^{-n/2} \exp(-n/2)$. When $\mu = \mu_0$, the MLE for σ^2 is $\sum (X_i - \mu_0)^2/n$, so the denominator is $(2\pi \sum (X_i - \mu_0)^2/n)^{-n/2} \exp(-n/2)$. Hence,

$$\Lambda_X(H_0, H_1) = \frac{(2\pi \sum (X_i - \bar{X})^2/n)^{-n/2}}{(2\pi \sum (X_i - \mu_0)^2/n)^{-n/2}} = \left[1 + \frac{n(\bar{X} - \mu_0)^2}{\sum (X_i - \bar{X})^2} \right]^{n/2}.$$

Thus, we reject H_0 when $T = \sqrt{n}(\bar{X} - \mu_0) / \sqrt{S_{XX}/(n-1)}$ is large.

Example 4.2 – Analysis of variance (ANOVA)

	Number of students	Total marks	Probability	Differential Equations	Vector Calculus
College A	44	318	36	29	32
College B	18	283	25	37	35
College C	16	294	28	35	37
College D	15	244	15	39	31
College E	7	280	24	34	35

We consider testing equality of means among $k > 2$ groups. Let n_i be the number of observations for the i -th group. Let $N = \sum_{i=1}^k n_i$. Our probability model is $X_{ij} = \mu_i + \varepsilon_{ij}$ for $j \in \{1, \dots, n_i\}$ and $i \in \{1, \dots, k\}$. We assume $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ are iid. The null hypothesis is $H_0 : \mu_1 = \dots = \mu_k$. The alternative hypothesis is $H_1 : (H_0 \text{ is not true})$.

We use a GLRT, with ratio of likelihoods being

$$\Lambda_X(H_0, H_1) = \frac{\sup_{\mu_1, \dots, \mu_k, \sigma^2} (2\pi\sigma^2)^{-N/2} \exp \left[-\sum_{i,j} \frac{(X_{ij} - \mu_i)^2}{2\sigma^2} \right]}{\sup_{\mu, \sigma^2} (2\pi\sigma^2)^{-N/2} \exp \left[-\sum_{i,j} \frac{(X_{ij} - \mu)^2}{2\sigma^2} \right]} = \left(\frac{S_0}{S_1} \right)^{N/2}$$

where $S_0 = \sum_{i,j} (X_{ij} - \bar{X}_{..})^2$ and $S_1 = \sum_{i,j} (X_{ij} - \bar{X}_{i.})^2$, and $\bar{X}_{..} = (1/N) \sum_{i,j} X_{ij}$ and $\bar{X}_{i.} = (1/n_i) \sum_{j=1}^{n_i} X_{ij}$. So, we should reject H_0 if S_0/S_1 is large. Note that

$$\begin{aligned} S_0 &= \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_{i.} + \bar{X}_{i.} - \bar{X}_{..})^2 \\ &= \sum_{i,j} \left[(X_{ij} - \bar{X}_{i.})^2 + 2(X_{ij} - \bar{X}_{i.})(\bar{X}_{i.} - \bar{X}_{..}) + (\bar{X}_{i.} - \bar{X}_{..})^2 \right] \\ &= \underbrace{\sum_{i,j} (X_{ij} - \bar{X}_{i.})^2}_{S_1} + \underbrace{\sum_i n_i (\bar{X}_{i.} - \bar{X}_{..})^2}_{\text{call this } S_2}. \end{aligned}$$

So, we reject H_0 if S_2/S_1 is large. We call S_1 within samples sum of squares, and S_2 between samples sum of squares. Note that $\sum_j (X_{ij} - \bar{X}_{i.})^2$ is independent of $\bar{X}_{i.}$, implying that S_1 is independent of S_2 . The latter is a function of $\{\bar{X}_{i.}\}_i$ and the observations are all independent for different i . Also, if H_0 is true, $S_1 \sim \sigma^2 \chi_{N-k}^2$ and $S_2 \sim \sigma^2 \chi_{k-1}^2$.

Definition 4.3 – F-distribution

If $V \sim \chi_n^2$ and $W \sim \chi_m^2$ are independent, we say that

$$F = \frac{V/n}{W/m}$$

has a F -distribution with (n, m) degrees of freedom, denoted $F_{n,m}$.

Thus, under H_0 , we have

$$\frac{S_2/(k-1)}{S_1/(N-k)} \sim F_{k-1, N-k}$$

and we obtain a size α test if we reject H_0 when the test statistic exceeds $F_{k-1, N-k}(\alpha)$.

Returning to the data, we have $k = 5$ and $N = 100$, so we consider $F_{4,95}$. At 5% significance level, we reject H_0 if the statistic $\geq F_{4,95}(0.05) \approx 2.47$. For the total marks, the F -statistic is 3.71 so we reject H_0 . The other F -statistics are 4.25, 1.70 and 0.50 respectively.

Recall the t -distribution, corresponding to $T = U/\sqrt{V/n}$ when $U \sim \mathcal{N}(0, 1)$ and $V \sim \chi_n^2$. Since $U^2 \sim \chi_1^2$, the statistic $T^2 = (U^2/1)/(V/n) \sim F_{1,n}$.

In Example 4.2, we described one-way ANOVA. Two-way ANOVA analyses data defined according to two variables. For example, a student's performance on exams depends on whether supervisions were given, and the monetary incentives offered. In this case, the model would be $Y_{ijk} = \alpha + \mu_j + \lambda_k + \varepsilon_{ijk}$ where (j, k) is a binary catalogue of supervisions/incentives. The test is for $H_0 : \mu_j = \lambda_k = 0$ versus $H_1 : \mu_j, \lambda_k \in \mathbb{R}$.

Suppose we have data $\{(X_1, Y_1), \dots, (X_n, Y_n)\}$ with $Y_i \in \mathbb{R}$ and $X_i \in \mathbb{R}^p$. In a linear model, we assume that

$$Y_i = \alpha + \beta_1(X_i)_1 + \beta_2(X_i)_2 + \dots + \beta_p(X_i)_p + \varepsilon_i$$

where α is the intercept, $(\beta_1, \dots, \beta_p)$ are the regression coefficients, and $(\varepsilon_1, \dots, \varepsilon_n)$ are random noise with mean 0 and variance σ^2 . We want to estimate $(\alpha, \beta_1, \dots, \beta_p)$. We call the X_i 's predictors or explanatory variables, and the Y_i 's response variables.

We often eliminate the intercept by adding a first predictor $(X_i)_1 = 1$ for all i , so β_1 plays the role of α . We can also model non-linear relationships. For example, if $X_{i1} = z_i$ and $X_{i2} = z_i^2$, then the "linear" model is $Y_i = \alpha + \beta_1 z_i + \beta_2 z_i^2 + \varepsilon_i$.

The coefficient β_j can be interpreted as the effect on the response of increasing X_{ij} by one unit while keeping all other predictors fixed. In general, however, linear models can only reveal correlations in data rather than causation.

We can formulate this with matrices. The linear model is

$$Y = X\beta + \varepsilon$$

where $Y = (Y_1 \dots Y_n)^\top$, $X = (X_{ij}) \in \mathbb{R}^{n \times p}$, $\beta = (\beta_1 \dots \beta_p)^\top$ and $\varepsilon = (\varepsilon_1 \dots \varepsilon_n)^\top$. We have the following assumptions.

- (i) $\mathbb{E}(\varepsilon) = 0$. That is, $\mathbb{E}(\varepsilon_i) = 0$ for all i , so $\mathbb{E}(Y_i) = X_{i1}\beta_1 + \dots + X_{ip}\beta_p$.
- (ii) $\text{Cov}(\varepsilon) = \sigma^2 I$. That is, $\text{var}(\varepsilon_i) = \sigma^2$ for all i , and $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$ for all $i \neq j$, so $\text{var}(Y_i) = \text{var}(x_i^\top \beta + \varepsilon_i) = \text{var}(\varepsilon_i) = \sigma^2$.

For now, we do not assume anything else about the distribution of the ε_i 's. Throughout our discussion, we assume that $X \in \mathbb{R}^{n \times p}$ is fixed, with $p \leq n$ and $\text{rank}(X) = p$.

Definition 4.4 – Least squares estimator

The least squares estimator $\hat{\beta}$ minimises the residual sum of squares

$$S(\beta) = \|Y - X\beta\|_2^2 = \sum_{i=1}^n (Y_i - x_i^\top \beta)^2$$

where $x_i^\top$ is the i -th row of X .

This is a quadratic polynomial in $(\beta_1, \dots, \beta_p)$ so $\hat{\beta}$ satisfies

$$\begin{aligned} \left[\frac{\partial S(\beta)}{\partial \beta_k} \right]_{\beta=\hat{\beta}} &= 0 \\ -2 \sum_{i=1}^n x_{ik} \left(Y_i - \sum_{j=1}^p x_{ij} \hat{\beta}_j \right) &= 0 \end{aligned}$$

for all $k \in \{1, \dots, p\}$. In matrix form, we have $X^\top X \hat{\beta} = X^\top Y$ so

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

Note that the least squares estimator is linear. That is, $\hat{\beta} = MY$ for some fixed matrix M . We can check that

$$(i) \quad \mathbb{E}(\hat{\beta}) = \mathbb{E}((X^\top X)^{-1} X^\top Y) = (X^\top X)^{-1} X^\top \underbrace{\mathbb{E}(Y)}_{=X\beta} = \beta \text{ so } \hat{\beta} \text{ is unbiased.}$$

$$(ii) \quad \text{Cov}(\hat{\beta}) = \text{Cov}((X^\top X)^{-1} X^\top Y) = (X^\top X)^{-1} X^\top \underbrace{\text{Cov}(Y)}_{=\sigma^2 I} X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}.$$

Theorem 4.1 – Gauss-Markov

Let $\beta^* = CY$ be any other unbiased linear estimator of β . Then,

$$\text{var}(t^\top \hat{\beta}) \leq \text{var}(t^\top \beta^*)$$

for any $t \in \mathbb{R}^p$.

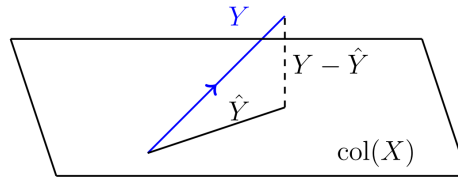
Proof. Note that the condition $\text{var}(t^\top \beta^*) - \text{var}(t^\top \hat{\beta}) = t^\top (\text{Cov}(\beta^*) - \text{Cov}(\hat{\beta}))t \geq 0$ for all $t \in \mathbb{R}^p$ is equivalent to $\text{Cov}(\beta^*) - \text{Cov}(\hat{\beta})$ being positive semidefinite. Let $A = C - (X^\top X)^{-1} X^\top$. Then, $\mathbb{E}(AY) = \mathbb{E}(\beta^*) - \mathbb{E}(\hat{\beta}) = 0$ since β^* and $\hat{\beta}$ are both unbiased. Since $\mathbb{E}(AY) = A\mathbb{E}(Y) = AX\beta$ for all $\beta \in \mathbb{R}^p$, this implies $AX = 0$. Furthermore,

$$\begin{aligned} \text{Cov}(\beta^*) &= \text{Cov}((A + (X^\top X)^{-1} X^\top)Y) \\ &= (A + (X^\top X)^{-1} X^\top) \underbrace{\text{Cov}(Y)}_{=\sigma^2 I} (A + (X^\top X)^{-1} X^\top)^\top \\ &= \sigma^2 (AA^\top + (X^\top X)^{-1} + \underbrace{AX(X^\top X)^{-1}}_{=0} + (X^\top X)^{-1} \underbrace{X^\top A^\top}_{=0}) \\ &= \sigma^2 AA^\top + \text{Cov}(\hat{\beta}). \end{aligned}$$

Thus, $\text{Cov}(\beta^*) - \text{Cov}(\hat{\beta}) = \sigma^2 AA^\top$ which is positive semidefinite. □

If we think about $(t_1, \dots, t_p)$ as the values of the predictors for a new sample, $t^\top \hat{\beta}$ and $t^\top \beta^*$ would be our prediction for the mean response using either estimator. The Gauss-Markov theorem says $t^\top \hat{\beta}$ always gives a smaller variance. Since $t^\top \hat{\beta}$ and $t^\top \beta^*$ are unbiased, the mean squared error is also smaller.

We can geometrically interpret the least squares method with the following.



Definition 4.5 – Orthogonal projection

We say that $P \in \mathbb{R}^{n \times n}$ is an orthogonal projection if it is idempotent ($P^2 = P$) and symmetric ($P^\top = P$).

Proposition 4.4

- (i) P is an orthogonal projection if and only if for all $v \in \text{col}(P)$, we have $Pv = v$, and for all $w \in \text{col}(P)^\perp$, we have $Pw = 0$. Equivalently, we can write $P = UU^\top$ where the columns of $U \in \mathbb{R}^{n \times k}$ form an orthonormal basis of $\text{col}(P)$, and $k = \text{rank}(P)$.
- (ii) If P is an orthogonal projection, then $I - P$ is also an orthogonal projection.
- (iii) $\|Y\|_2^2 = \|PY\|_2^2 + \|(I - P)Y\|_2^2$ for all $Y \in \mathbb{R}^n$.
- (iv) The eigenvalues of P are all 0 or 1. In particular, $\text{tr}(P) = \text{rank}(P)$.

The fitted values in linear regression are the predicted mean responses $\hat{Y} = X\hat{\beta}$, and the residues are $Y - \hat{Y}$. Note that $\hat{Y} = \underbrace{X(X^\top X)^{-1}X^\top}_{:=P} Y$ and $Y - \hat{Y} = (I - P)Y$.

Proposition 4.5

P is the orthogonal projection onto $\text{col}(X)$.

Proof. P is clearly symmetric and $P^2 = [X(X^\top X)^{-1}X^\top][X(X^\top X)^{-1}X^\top] = X(X^\top X)^{-1}X^\top = P$, so P is an orthogonal projection. Also, $\text{col}(P) = \text{col}(X)$ since for any $a \in \mathbb{R}^n$, we have $Pa = X(X^\top X)^{-1}X^\top a \in \text{col}(X)$, and conversely, if $b = Xc \in \text{col}(X)$, then $b = Xc = X(X^\top X)^{-1}X^\top Xc = Pb \in \text{col}(P)$. $\square$

Corollary 4.2

The vector of fitted values $\hat{Y} = PY$ is the orthogonal projection of Y onto $\text{col}(X)$, and the vector of residues $Y - \hat{Y} = (I - P)Y$ is the orthogonal projection of Y onto $\text{col}(X)^\perp$.

In addition to $\mathbb{E}(\varepsilon) = 0$ and $\text{Cov}(\varepsilon) = \sigma^2 I$, we will now assume that $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$. We wish to estimate the parameters (β, σ^2) . The observations of Y have pdf

$$f_Y(Y | \beta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - X_i^\top \beta)^2 \right].$$

The log-likelihood is

$$l(\beta, \sigma^2) = \text{const.} - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \underbrace{\|Y - X\beta\|_2^2}_{S(\beta)}.$$

Note that for any $\sigma^2 > 0$, $l(\beta, \sigma^2)$ is maximised when $S(\beta)$ is minimised. Thus, the MLE of β is the least

squares estimator $\hat{\beta}$. The MLE of σ^2 minimises $l(\hat{\beta}, \sigma^2)$. We need

$$\frac{\partial l(\hat{\beta}, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} S(\hat{\beta}) = 0$$

$$\hat{\sigma}^2 = \frac{S(\hat{\beta})}{n} = \frac{\|Y - X\hat{\beta}\|_2^2}{n} = \frac{\|Y - \hat{Y}\|_2^2}{n} = \frac{\|(I - P)Y\|_2^2}{n}.$$

Theorem 4.2

If $X \sim \mathcal{N}(0, \sigma^2 I)$ and P is an orthogonal projection matrix, then

- (i) $PX \sim \mathcal{N}(0, \sigma^2 P)$ and $(I - P)X \sim \mathcal{N}(0, \sigma^2(I - P))$ are independent;
- (ii) $\|PX\|_2^2 / \sigma^2 \sim \chi_{\text{rank}(P)}^2$.

Proof. (i) is on the example sheets. For (ii), note that

$$\frac{\|PX\|_2^2}{\sigma^2} = \frac{(PX)^\top PX}{\sigma^2} = \frac{X^\top (UU^\top)^\top (UU^\top) X}{\sigma^2}$$

where the columns of U form a basis for $\text{col}(P)$. This is then equal to

$$\frac{\|U^\top X\|_2^2}{\sigma^2} = \sum_{i=1}^{\text{rank}(P)} \frac{(U^\top X)_i^2}{\sigma^2}.$$

But $U^\top X \sim \mathcal{N}(0, \sigma^2 U^\top U) = \mathcal{N}(0, \sigma^2 I)$, so $(U^\top X)_i / \sigma$ are iid $\mathcal{N}(0, 1)$ for all i . This gives the result. $\square$

Corollary 4.3

- (i) $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(X^\top X)^{-1})$.
- (ii) $(n/\sigma^2)\hat{\sigma}^2 \sim \chi_{n-p}^2$.
- (iii) $\hat{\beta}$ and $\hat{\sigma}^2$ are independent.

Proof. For (i), we already know $\mathbb{E}(\hat{\beta}) = \beta$ and $\text{Cov}(\hat{\beta}) = \sigma^2(X^\top X)^{-1}$. Since $\hat{\beta} = (X^\top X)^{-1}X^\top Y$ is linear in $Y \sim \mathcal{N}(X\beta, \sigma^2 I)$, we know that $\hat{\beta}$ is also multivariate normal.

For (ii), note that

$$\frac{n\hat{\sigma}^2}{\sigma^2} = \frac{\|(I - P)Y\|_2^2}{\sigma^2} = \frac{\|(I - P)(X\beta + \varepsilon)\|_2^2}{\sigma^2} = \frac{\|(I - P)\varepsilon\|_2^2}{\sigma^2} \sim \chi_{\text{tr}(I-P)}^2$$

by Theorem 4.1 (ii) and Proposition 4.4 (iv). Note that $\text{tr}(I - P) = \text{tr}(I) - \text{tr}(P) = n - \text{rank}(P) = n - p$ since $\dim(\text{col}(P)) = \dim(\text{col}(X)) = p$ as X is full rank.

For (iii), note that $\hat{\sigma}^2$ is a function of $(I - P)\varepsilon$ and

$$\hat{\beta} = (X^\top X)^{-1}X^\top Y = (X^\top X)^{-1}X^\top (X\beta + \varepsilon) = \beta + (X^\top X)^{-1}X^\top \varepsilon = \beta + (X^\top X)^{-1}X^\top P\varepsilon$$

so $\hat{\beta}$ is a function of $P\varepsilon$. By Theorem 4.1 (i), $\hat{\beta}$ and $\hat{\sigma}^2$ are independent. $\square$

Suppose we want a $100(1 - \alpha)\%$ confidence interval for the coefficients in the linear model. Without loss of

generality, consider β_1 . Note that

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2(X^\top X)^{-1}_{11}}} \sim \mathcal{N}(0, 1) \text{ is independent of } \frac{n\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-p}^2.$$

We use this to construct a pivot

$$\frac{\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2(X^\top X)^{-1}_{11}}}}{\sqrt{\frac{\hat{\sigma}^2 n}{\sigma^2(n-p)}}} \sim \frac{\mathcal{N}(0, 1)}{\sqrt{\frac{\chi_{n-p}^2}{n-p}}} \sim t_{n-p}.$$

Rearranging, we arrive at the confidence interval

$$\hat{\beta}_1 \pm t_{n-p} \left(\frac{\alpha}{2} \right) \sqrt{\frac{(X^\top X)^{-1}_{11} \hat{\sigma}^2 n}{n-p}}.$$

Suppose we wish to test $H_0 : \beta_1 = 0$ (the first predictor has no effect on the response) versus $H_1 : \beta_1 \neq 0$. By the duality of confidence intervals and hypothesis tests, we can obtain a test of size α for H_0 by defining the acceptance region $A = \{\beta : 0 \in I(\beta)\}$. In other words, reject H_0 if

$$|\hat{\beta}_1| > t_{n-p} \left(\frac{\alpha}{2} \right) \sqrt{\frac{(X^\top X)^{-1}_{11} \hat{\sigma}^2 n}{n-p}}.$$

There is nothing special about testing $\beta_1 = 0$. We could just as easily derive a test of size α for $H_0 : \beta_1 = \beta'_1$ for any $\beta'_1 \in \mathbb{R}$. We could also determine a confidence set (which is an ellipsoid for this case) for the whole vector of coefficients β by constructing a pivot based on $\hat{\beta} - \beta \sim \mathcal{N}(0, \sigma^2(X^\top X)^{-1})$.

Suppose we have a linear model $Y = X\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$. We wish to test the null hypothesis that a whole collection of predictors has no effect on the response. Without loss of generality, take the first $p_0 \leq p$ coefficients. So, $H_0 : \beta_1 = \beta_2 = \dots = \beta_{p_0} = 0$ versus $H_1 : (\beta_1, \dots, \beta_{p_0}) \in \mathbb{R}^{p_0}$. We write $X = (X_0 \ X_1)$ where $X_0 \in \mathbb{R}^{n \times p_0}$ and $X_1 \in \mathbb{R}^{n \times (p-p_0)}$. Write

$$\beta = \begin{pmatrix} \beta^0 \\ \beta^1 \end{pmatrix} \quad \text{where } \beta^0 = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_{p_0} \end{pmatrix} \text{ and } \beta^1 = \begin{pmatrix} \beta_{p_0+1} \\ \vdots \\ \beta_p \end{pmatrix}.$$

The null model has $\beta^0 = 0$, so it is a linear model $Y = X\beta + \varepsilon = X_1\beta^1 + \varepsilon$. We will write $P = X(X^\top X)^{-1}X^\top$ and $P_1 = X_1(X_1^\top X_1)^{-1}X_1^\top$. Since X and P have full rank, so does X_1 and P_1 . Recall that the MLE in the linear model is

$$\begin{aligned} \sup_{\substack{\beta \in \mathbb{R}^p \\ \sigma^2 > 0}} l(\beta, \sigma^2) &= l(\hat{\beta}, \hat{\sigma}^2) = -\frac{n}{2} \log(\hat{\sigma}^2) - \underbrace{\frac{S(\hat{\beta})}{2\hat{\sigma}^2}}_{=n/2} + \text{const.} \\ &= -\frac{n}{2} \log \left[\frac{\|(I-P)Y\|_2^2}{n} \right] + \text{const.} \end{aligned}$$

Thus, a GLRT for $H_0 : \beta^0 = 0$ versus $H_1 : \beta^0 \in \mathbb{R}^{p_0}$ computes

$$\sup_{\substack{\beta \in \mathbb{R}^p \\ \sigma^2 > 0}} l(\beta, \sigma^2) - \sup_{\substack{\beta^0 = 0 \\ \beta^1 \in \mathbb{R}^{p-p_0} \\ \sigma^2 > 0}} l(\beta, \sigma^2) = \frac{n}{2} \left(-\log \left[\frac{\|(I-P)Y\|_2^2}{n} \right] + \log \left[\frac{\|(I-P_1)Y\|_2^2}{n} \right] \right).$$

The GLRT rejects H_0 when this is large. This expression is monotone in

$$\frac{\|(I-P_1)Y\|_2^2}{\|(I-P)Y\|_2^2} = \frac{\|(I-P+P-P_1)Y\|_2^2}{\|(I-P)Y\|_2^2} = \frac{\|(I-P)Y\|_2^2 + \|(P-P_1)Y\|_2^2 + 2Y^\top(I-P)(P-P_1)Y}{\|(I-P)Y\|_2^2}.$$

Lemma 4.1

We have $(I - P)(P - P_1) = 0$. Furthermore, $P - P_1$ is an orthogonal projection with rank p_0 .

Thus, the test statistic is monotonically increasing in

$$\frac{\|(P - P_1)Y\|_2^2 \frac{1}{p_0}}{\|(I - P)Y\|_2^2 \frac{1}{n-p}} =: F.$$

Under H_0 , we have $(P - P_1)Y = (P - P_1)(X\beta + \varepsilon) = (P - P_1)(X_1\beta^1 + \varepsilon) = (P - P_1)\varepsilon$. Thus,

$$F = \frac{\|(P - P_1)\varepsilon\|_2^2 \frac{1}{p_0}}{\|(I - P)\varepsilon\|_2^2 \frac{1}{n-p}}.$$

Observe that $\|(P - P_1)\varepsilon\|_2^2 / \sigma^2 \sim \chi_{p_0}^2$ and $\|(I - P)\varepsilon\|_2^2 / \sigma^2 \sim \chi_{n-p}^2$. Also, the vector $\begin{pmatrix} (P - P_1)\varepsilon \\ (I - P)\varepsilon \end{pmatrix}$ is normal and $\text{Cov}((P - P_1)\varepsilon, (I - P)\varepsilon) = 0$ by Lemma 4.1. Thus, $F \sim F_{p_0, n-p}$. A GLRT, or F -test of size α , rejects H_0 when

$$\frac{\|(P - P_1)Y\|_2^2 \frac{1}{p_0}}{\|(I - P)Y\|_2^2 \frac{1}{n-p}} > F_{p_0, n-p}(\alpha).$$

Suppose we observe data $\{(X_1, Y_1), \dots, (X_n, Y_n)\}$ with $X_i \in \mathbb{R}^p$ and $Y_i \in \{0, 1\}$. We wish to predict Y_i from X_i . The logistic model is

$$p(x_i) = \mathbb{P}(Y_i | x_i) = \frac{\exp(\beta^\top x_i)}{1 + \exp(\beta^\top x_i)}$$

where $\beta \in \mathbb{R}^p$ is the vector of parameters we want to estimate. The right hand side is called the sigmoid function of $\beta^\top x$. Rearranging, the model is

$$\log \left(\frac{p(x)}{1 - p(x)} \right) = \beta^\top x.$$

The likelihood can be written as

$$L(\beta) = \prod_{i: y_i=1} p(x_i) \prod_{i: y_i=0} (1 - p(x_i)) = \prod_{i=1}^n p(x_i)^{y_i} (1 - p(x_i))^{1-y_i}$$

and we wish to maximise

$$l(\beta) = \sum_{i=1}^n [y_i \log(p(x_i)) + (1 - y_i) \log(1 - p(x_i))].$$

Based on an estimate $\hat{\beta}$ of the logistic coefficients, we can compute $\hat{p}(x) = \exp(\hat{\beta}^\top x) / (1 + \exp(\hat{\beta}^\top x))$ on a new input, and classify x according to $\mathbb{1}\{\hat{p}(x) > 1/2\}$.